

What LLM Forecasters Reveal When You Ask Them About Their Uncertainty: A Per-Family, Cross-Corpus Study

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Abstract

LLM forecast calibration is usually reported at the panel-aggregate stratum, which hides where the per-model signal actually lives. We elicit five uncertainty primitives in the same prompt (worry scalar, bid-ask spread, self-predicted Brier interval, reasoning trajectory, reference-class outside view) from five families (Claude Opus 4.7, GPT-5.5, GPT-5.4-mini, Gemini 2.5 Flash, DeepSeek Chat) on a public-domain $N=42$ corpus, with every claim resolved through three legal verdicts (**h1_supported**, **h0_kept**, **inconclusive_underpowered**) and required sample sizes pre-computed via Fisher- z power. Three claims survive: (1) the per-model worry-Brier relationship is direction-flipped between families and replicates across three independent pilots ($N=264$); (2) a multi-channel OLS on the public corpus generalizes for one of five families under leave-one-out cross-validation (GPT-5.5, $R^2_{\text{LOO}}=+0.31$; the other four are at or below zero); (3) a reference-class outside view, the model first lists five similar resolved questions and emits their historical YES rate before its own probability, partially replicates cross-corpus, with $\rho=+0.39$ **h1_supported** for GPT-5.4-mini, beating that family’s own point estimate ($\rho=+0.21$). The bid-ask spread and self-predicted-Brier interval are introduced as novel elicitation primitives. The audit pipeline retracted four prior-program “null” findings to **inconclusive_underpowered** and five cross-corpus claims to corpus-specific; we report the retractions transparently. The deployable recipe is a per-family elicitation policy, not panel-averaged probability.

1 Introduction

Eigenproblem. Which LLM forecasting behaviours are structural inheritances of the training-text distribution the model samples from, and along which axes can the transfer pattern be predicted ex ante from the structural origin of each behaviour?

Epistemic shape. The paper is empirical first and theoretical second. We ran a multi-pilot forecast-calibration program (five-model panel, two corpora, 30+ measured findings, pre-registration discipline) without a unifying theoretical commitment. The three-axis inheritance frame below was **induced from the body of measurements in retrospect** as the most parsimonious structure organising the per-family heterogeneity, the universal patterns, and the bias-transfer findings together. We treat the framework’s status accordingly: it earns the right to be called a unifying frame only to the extent that it makes pre-registered predictions on bias transfers we had not yet measured at the time of synthesis. Section §15 reports the in-sample evidence; section §17 reports the predictive test on a held-out slate of biases pre-registered under the framework.

The three-axis inheritance frame (inductively derived). The findings in this paper are organised by three structural axes along which training-text inheritance shows up in the forecast distribution. Each axis governs a different family of measurements; together they cover the program’s per-family heterogeneity, universal patterns, and bias-transfer findings.

- **Axis 1, Elicitation surface.** Which channels the model exposes when asked to forecast (a point estimate p_{success} , a bid-ask spread $p_{\text{buy yes max}}/p_{\text{sell yes min}}$, a tail-worry scalar, a self-predicted Brier interval, an asymmetric-loss decision threshold). The model emits the channels that its training distribution rewards; channels differ across families because training mixes differ.
- **Axis 2, Bias-mechanism class.** A three-cell taxonomy on which human cognitive biases transfer to LLM forecasting and how. *Pure heuristic* biases (cognitive shortcuts with high natural-speech footprint, like anchoring or self-overconfidence) INHERIT. *Pure motivational* biases (intrinsic utility / fear / consumption preference, like loss aversion or hyperbolic discounting) ESCAPE: the model has no utility function, the text has no representation of the mechanism, no transfer. *Systemic motivational* biases (intrinsic utility but heavy case-study representation in training text, like sunk cost or in-group framing) MIMIC: the model emits utility-like behaviour because the text frequency of the bias being discussed drives the output, not a utility function operating.
- **Axis 3, Alignment overlay.** RLHF / alignment damps a subset of inherited biases and reshapes per-family channel surfaces. Per-family heterogeneity in worry-channel sign, channel decomposition R^2 , and meta-cognitive routing all live on this axis.

The contribution shape is the **synthesis** of the three axes plus the MIMIC category as a specifically novel articulation on axis 2: prior LLM-bias work has not separated transfer-because-mechanism-is-shared (inherit) from transfer-because-bias-is-discussed-extensively-in-text (mimic). The MIMIC category lets the same surface observation carry different interpretations depending on which cell the bias belongs to, and predicts distinct signatures under explicit anti-bias prompting.

Methodological backbone. A power-aware verdict resolver (Fisher- z sample-size computation before fire, three legal verdicts after `h1_supported` / `h0_kept` / `inconclusive_underpowered`, equivalence testing instead of “ $p > 0.05$ ” as a no-effect claim, BH-FDR across panel tests, leave-one-out R^2 at small N) sits underneath every empirical claim in the paper. Internally, 8 of 12 prior nulls in this program turned into `inconclusive_underpowered` under the resolver, and 5 cross-corpus claims retracted to corpus-specific. The discipline ITSELF is a methodological contribution because the same overcalling failure mode plausibly affects the field’s existing $N=5-20$ cross-corpus headlines. A second-order application (corpus-validity drift: when a benchmark’s resolution dates are post-cutoff for the publication-era LLMs and pre-cutoff for the next-generation LLMs replicating against it) is the most consequential field-corrective the discipline has produced, applied to the Halawi 2024 dataset [3] in §18.

Reading order. Axes 1–3 are introduced in §3; Axis 1 findings sit in §7, §8, §11; Axis 2 findings sit in §15 (with the MIMIC category as the load-bearing original contribution) and inside the universal-patterns deployable rules in §14 (the confident-NO discount as the INHERIT cell in deployment shape); Axis 3 findings sit in §5, §10, and the per-family channel-routing results. The methodological backbone runs underneath all of them and is collected in §18. An honest scope statement (§19) and the reproducibility plan (§20) close the paper.

2 Setup

2.1 Five LLM families

Claude Opus 4.7 (Anthropic, via Claude Code CLI), GPT-5.5 and GPT-5.4-mini (OpenAI, via Codex CLI), Gemini 2.5 Flash (Google GenAI API), DeepSeek Chat (DeepSeek API). All API calls temperature=0 unless explicitly noted. All paths audited for web-search-tool access; all prompts include an explicit “do NOT use web search” instruction.

2.2 Two corpora, four confounded axes

We use a *low-overlap corpus* of $N=15$ bespoke questions drawn from a research-internal domain with effectively zero LLM-pretraining overlap (formal-proof tactic combinations, gated state transitions in a substrate-specific apparatus, ground-truth-confirmed forecasting contracts), and a *public-domain corpus* of $N=42$ contracts from prediction markets (Manifold, Polymarket), stock-close thresholds, and ETF moves.

Axis	Low-overlap ($N=15$)	Public-domain ($N=42$)
LLM pre-training overlap	Effectively zero	Substantial
Question length	247 chars (35 words)	99 chars (17 words)
Question structure	Multi-conditional, code/math symbols, parenthetical embedding	Atomic, simple syntactic structure
Implicit base rate	Ill-defined (bespoke operations)	Public-market priors well-trained

The four axes are confounded in the current data. Low-overlap-corpus questions are long-complex-alien-no-public-prior *together*; public-domain are short-simple-familiar-with-public-prior *together*. Our cross-corpus result (§9) is therefore evidence that *at least one* of these axes matters; a 4-cell de-confounded corpus design that breaks the confound is the highest-priority methodological follow-up (§19).

The low-overlap corpus is research-internal and not publicly releasable in raw form; the reproducibility plan (§20) addresses this through a sanitized release with neutral identifiers and a parallel public-but-niche-academic corpus matching the four-axis profile.

2.3 Pre-registration and power-aware verdicts

Every pilot is pre-registered in a SQLite `pre_registrations` table before launch with hypothesis, null, target effect size, n_{required} (auto-computed via Fisher- z at $\alpha=0.05$ two-tailed, 80% power, Spearman correction +6%), falsifiers, and pass-gate. Resolution uses three legal verdicts:

- **h1_supported**: 95% CI on observed ρ excludes 0.
- **h0_kept**: 95% CI wholly within $(-\text{target}, +\text{target})$; data rules out a meaningful effect.
- **inconclusive_underpowered**: observed $|\rho|$ below detectability at the run N ; CI wide.

Brier-delta tests use paired-contract sign-flip permutation with a 90% bootstrap CI plus a BIC-approximation Bayes factor. “No effect” claims also require TOST equivalence at a pre-stated bound. Power thresholds at $\alpha=0.05$ / power= 0.80: $N=15 \Rightarrow |\rho| \geq 0.68$; $N=42 \Rightarrow 0.43$; $N=91 \Rightarrow 0.30$.

3 The three-axis inheritance frame

Before reporting the empirical findings, we name the structural axes the program’s measurements turned out to organise along. These axes were not stated in advance; they are the most parsimonious reading of the body of findings the program produced. The framework’s predictive content is tested separately (§17).

Axis 1: elicitation surface. An LLM forecaster, when asked, emits a set of channels: at minimum a point estimate p_{success} ; optionally a tail-worry scalar (“how worried are you?”), a bid-ask spread ($p_{\text{buy yes max}}, p_{\text{sell yes min}}$), a self-predicted Brier interval ($b_{\text{lo}}, b_{\text{hi}}$), an asymmetric-loss decision threshold under specified false-positive vs. false-negative cost regimes, or an outside-view base rate from analogous prior contracts. The set of channels the model surfaces is itself the first inheritance signature: training-mix differences across model families produce different elicitation surfaces, even when prompted identically. §7 introduces the bid-ask spread as a novel runtime calibration warning that is elicitable on the low-overlap corpus; §8 shows the three uncertainty channels are statistically independent on a paired panel, requiring per-family channel routing; §11 shows only one family generalises a multi-channel decomposition under leave-one-out cross-validation. All three findings sit on this axis.

Axis 2: bias-mechanism class. We propose a three-cell taxonomy for how human cognitive biases transfer to LLM forecasting.

- **PURE HEURISTIC** (cognitive shortcut, high natural-speech footprint). Examples: anchoring [12], self-overconfidence [6], the gambler’s fallacy [11]. Predicted manifestation: **inherit** (training-text frequency drives the LLM output directly via the token distribution).
- **PURE MOTIVATIONAL** (intrinsic utility / fear / consumption preference; low representation in raw analytical text). Examples: loss aversion in the Kahneman-Tversky sense [4], cumulative prospect theory probability weighting [13], hyperbolic discounting [2]. Predicted manifestation: **escape** (the LLM has no utility function and the text has no representation of the bias mechanism, so no transfer).
- **SYSTEMIC MOTIVATIONAL** (intrinsic utility BUT heavy case-study / behavioural-economics-literature representation in training text). Examples: sunk-cost reasoning [1], endowment / status-quo bias [10], in-group framing [8]. Predicted manifestation: **mimic** (the LLM emits utility-like behaviour, but the mechanism producing the output is text-frequency representation of the bias being discussed in training text, not a utility function operating).

The MIMIC cell is the load-bearing original contribution of the framework. Prior LLM-bias work has not separated transfer-because-the-mechanism-is-shared (INHERIT) from transfer-because-the-bias-is-discussed-extensively-in-text (MIMIC). The cell distinction matters: MIMIC predicts that explicit anti-bias prompting collapses the effect (the corrective narrative is also present in training text and can be retrieved), where INHERIT predicts the effect persists under anti-bias prompting because it is the model’s default sampling pattern, not a retrievable narrative.

Axis 3: alignment overlay. RLHF and post-training alignment damp some inherited biases (truthfulness, helpfulness training reduces obvious factual hallucinations and overt in-group preference) and reshape per-family channel surfaces. Per-family worry-Brier sign differences (§5), per-family channel-decomposition R^2 differences (§11), and the meta-cognitive routing failure where

three of five families pick the WRONG channel they should rely on at $p \leq 0.005$ each (§12) all live on this axis. The empirical signature is that family identity, conditional on channel and contract, is itself a load-bearing predictor.

The methodological backbone is orthogonal to the three axes. A verdict-resolver discipline (Fisher- z power calculation before fire, three legal verdicts after, equivalence testing, BH-FDR, LOO- R^2 at small N) sits underneath every axis-1/2/3 measurement. The discipline produced the program’s most consequential retraction: 8 of 12 prior “no effect” claims turned into `inconclusive_underpowered` under the resolver, and the corpus-validity-drift filter (`resolve_date > max(panel_cutoff)`) empties the most-cited 2024 LLM-forecasting benchmark for the current LLM generation (§18).

4 Foundation: the worry scalar is Bayesian-coherent tail risk

The basic uncertainty primitive used throughout this paper is `tail_insurance_premium` (worry scalar, 1–100, requested in the same prompt as the binary forecast). This primitive was validated in a separate within-program pilot before any of the downstream analyses: at pool level $N=590$ on an independent cross-pool forecast corpus, the worry scalar is a Bayesian-coherent tail-risk signal, more-worried forecasts ARE more likely to be wrong, in expectation. This is the only well-powered *positive* pool-level validation of a verbalized-uncertainty primitive in our program; everything that follows uses worry as a base channel, and the per-model decomposition in §5 shows the pool-level signal is the average of opposite per-family signs.

5 Per-family worry-Brier direction split, three-pilot replication

For each LLM family we compute the rank correlation between the worry score and the realized squared error (Brier) of the forecast that the worry accompanied. Three independent pilots ($N=164$, $N=85$, and $N=115$; cumulative $N=264$) replicate this analysis:

- Claude Opus 4.7: $\rho < 0$, most worried when most right (anti-Dunning-Kruger).
- GPT-5.5: $\rho < 0$, same direction.
- GPT-5.4-mini: $\rho > 0$, direction-sensible (worried when wrong).

Per-pilot detectable $|\rho|$ at 80% power is ≈ 0.30 for $N=85$; observed $|\rho|$ values span 0.31–0.56 across families, clearing detectability for Claude and GPT-5.5 individually. Cross-pilot Stouffer Z meta-analysis combines the three pilots at combined $p < 0.01$. **This is the per-model finding that survives the largest paired-permutation budget in the program.**

A natural reviewer hypothesis is that the sign-flip is mediated by task complexity: complex tasks produce longer rationales; longer rationales correlate with both higher stated worry AND lower Brier (more thorough reasoning); the apparent inversion is an artifact of this joint mediation. We test the mediator directly on $N=45\text{--}91$ per family. None of the five families exhibits the predicted pattern of “long rationale $\rightarrow$ high worry AND long rationale $\rightarrow$ low Brier.” Claude’s $\rho(\text{rat_length}, \text{tip}) \approx 0$ (claude does not worry more about long rationales); Gemini’s $\rho(\text{rat_length}, \text{Brier})$ is POSITIVE, the opposite of the reviewer’s predicted direction. **The task-complexity-via-rationale-length mediator is rejected.**

The refined mechanism we propose is given in §10 after §6 provides the upstream decoupling result.

6 Cognitive content and calibration shift are decoupled

Across three pilots (pooled $N=320$ analytic), we measure the within-family Spearman correlation between the count of failure-mode words in the rationale and the per-row Brier:

Family	N	$\rho(\text{failure-mentions, Brier})$
Claude Opus 4.7	≈ 60	+0.27
GPT-5.5	≈ 60	+0.18
GPT-5.4-mini	≈ 60	+0.21
Gemini 2.5 Flash	60	−0.045
DeepSeek Chat	59	+0.049

At $N=60$ per family the detectable $|\rho|$ at 80% power is ≈ 0.36 . Gemini and DeepSeek’s near-zero ρ values fall within the **h0_kept** band at $|\rho|<0.20$. The model produces failure-mode reasoning text on demand, cognitive shift is real, but does not recompute the point estimate from that content. This is the cleanest mechanism evidence in the program and connects to the chain-of-thought-faithfulness lineage [5]; the original trio shows weak positive correlation consistent with partial propagation rather than full decoupling.

7 Bid-ask spread: a novel meta-uncertainty primitive

Instead of asking the model for one probability, we elicit three in the same prompt: the highest probability at which it would BUY a YES contract, its point estimate, and the lowest probability at which it would SELL YES. The spread (sell − buy) is the model’s self-reported uncertainty about its own probability, a Knightian bracket distinct from the unidirectional worry scalar.

Family	Mean spread	$\rho(\text{spread, Brier})$	95% CI (Fisher- z)	Verdict
Claude Opus 4.7	0.21	+0.49	[−0.03, +0.80]	CI barely touches 0
GPT-5.5	0.24	−0.05	[−0.55, +0.47]	inconclusive
GPT-5.4-mini	0.26	+0.69	[+0.28, +0.89]	h1_supported, $p=0.013$
Gemini 2.5 Flash	0.21	+0.40	[−0.15, +0.75]	inconclusive
DeepSeek Chat	0.26	+0.69	[+0.27, +0.89]	h1_supported, $p=0.013$

Pre-registered pass-gate met: two of five families clear $\rho > 0.3$ with non-zero-crossing CI at $N=15$ low-overlap. No family shows the wrong sign. To our knowledge no prior LLM-forecasting paper has elicited bid-ask spread as an uncertainty primitive. The market-microstructure framing forces the model into a more demanding cognitive task than the scalar worry, which we conjecture is exactly why it produces a more honest meta-uncertainty signal (the elaboration is supported by the channel-orthogonality result in §8).

Cross-corpus failure-to-replicate. On the public-domain $N=42$ corpus, all per-family CIs cross zero (detection threshold $|\rho|\geq 0.43$ at 80% power for $N=42$). The point-estimate ρ values for GPT-5.4-mini (−0.11) and DeepSeek (−0.07) flip sign. Either (a) the primitive is corpus-specific consistent with the four-axis confound, or (b) $N=42$ remains underpowered to confirm a smaller-but-real positive effect. To distinguish requires $N\geq 91$ public-domain. The honest framing is *low-overlap-corpus positive with explicit cross-corpus failure-to-replicate*.

8 LLM uncertainty channels are statistically independent

Pooling three low-overlap-corpus full pilots (worry-scalar baseline, bid-ask spread, reasoning-trajectory variance) on the $N=75$ rows where all three are observed, the pairwise Pearson correlation matrix among the three uncertainty channels:

	worry scalar	bid-ask spread	trajectory variance
worry scalar	1.00	+0.08	+0.17
bid-ask spread	+0.08	1.00	+0.03
trajectory variance	+0.17	+0.03	1.00

The off-diagonals are all below 0.20. PCA would not produce dimensionality reduction. Each primitive is measuring something genuinely different about how a model expresses uncertainty.

Per-family $\rho(\text{channel}, \text{Brier})$ reveals that each model family has a different “best channel”:

Family	$\rho(\text{worry}, \text{Brier})$	$\rho(\text{spread}, \text{Brier})$	$\rho(\text{traj_var}, \text{Brier})$	Best channel
Claude Opus 4.7	−0.38 (sign-flipped)	+0.02	+0.08	worry (with sign flip)
GPT-5.5	+0.07	+0.12	−0.18	weak across all
GPT-5.4-mini	−0.18	+0.55	−0.22	bid-ask spread
Gemini 2.5 Flash	+0.10	+0.10	+0.25	trajectory variance
DeepSeek Chat	−0.26 (sign-flipped)	+0.29	−0.10	bid-ask spread

There is no universal best uncertainty primitive across LLMs. The deployable recipe is a per-family elicitation policy: identify the best channel per family via a pre-registered pilot, then use that channel as the runtime calibration warning. Per-primitive per-family single-channel R^2 on the same-pilot Brier is 10–26% for the families where the channel works (GPT-5.4-mini’s bid-ask explains 26% of bid-ask-pilot Brier; DeepSeek’s 19%). The multi-channel R^2 from the all-channels-in-one-prompt pilot is reported with leave-one-out cross-validation in §11, where only GPT-5.5 survives LOO on the public-domain corpus.

Scope of the orthogonality claim. The result is at the *expression* level. A reasoning-decomposition claim would require mechanistic interpretability tooling, logit lens, attention probing, causal intervention on chain-of-thought spans, which we do not use. Our primitives are black-box behavioral probes of uncertainty expression; the deployment implication (per-family routing) is unchanged but the mechanistic-reasoning claim is left for white-box work.

9 Frequency framing transfers in a corpus-specific way

Tetlock-style “imagine 10 similar contracts; how many resolve as success?” [9] framing on the low-overlap corpus ($N=15$ each, paired permutation vs the same-corpus baseline):

Family	$\Delta\text{-Brier}$	95% CI	p (perm)	Verdict
Claude Opus 4.7	−0.006	[−0.05, +0.03]	0.85	inconclusive
GPT-5.5	+0.014	[−0.06, +0.09]	0.74	inconclusive
GPT-5.4-mini	−0.118	[−0.21, −0.04]	0.013	h1 supported
Gemini 2.5 Flash	−0.118	[−0.28, +0.03]	0.19	inconclusive
DeepSeek Chat	+0.115	[−0.02, +0.28]	0.18	inconclusive

On the public-domain $N=42$ corpus:

Family	Δ -Brier	95% CI	Verdict at bound 0.05
Claude Opus 4.7	-0.003	$[-0.030, +0.028]$	h0_kept
GPT-5.5	+0.005	$[-0.012, +0.023]$	h0_kept
GPT-5.4-mini	+0.003	$[-0.042, +0.053]$	borderline
Gemini 2.5 Flash	+0.052	$[+0.000, +0.112]$	not h0_kept ($p=0.071$)
DeepSeek Chat	-0.001	$[-0.031, +0.030]$	h0_kept

GPT-5.4-mini’s low-overlap-corpus improvement does NOT replicate on the public-domain corpus. Direction goes from -0.118 to $+0.003$. For 3 of 5 families the public-corpus Brier-delta is **h0_kept** at bound ± 0.05 , the data actively rules out a 5%-Brier effect on public-domain questions. Sub-source stratification within the public corpus reveals heterogeneity the aggregated null masks: Gemini on Polymarket-style questions trends $\Delta = +0.138$ ($p=0.14$); DeepSeek shows opposite-sign effects on Manifold-bulk vs Polymarket; GPT-5.4-mini is genuinely corpus-specific (every sub-source $\Delta \in [-0.002, +0.007]$, no false-negative from aggregation).

The clean claim. Frequency framing’s effect on LLM forecasters is corpus-specific and model-specific: it improves GPT-5.4-mini on low-overlap apparatus questions only. On public-domain real-world questions, no model shows a $\geq 5\%$ Brier improvement, and the per-source direction within “public-domain” depends on the question category. Pre-registration prevented us from over-stating the cross-corpus claim as a general failure-of-transfer; the specific finding is the corpus-dependent transfer.

10 Refined mechanism for the worry-scalar sign-flip: shared topic-trigger $\times$ per-family accuracy

A reviewer-flagged concern is that the affect-proxy hypothesis (§5 early framing) was still just a hypothesis. We ran four mechanism probes on existing within-program data to disambiguate.

Probe 1. Within-family rank correlation between (across-family p -variance per contract, used as a proxy for intrinsic question difficulty) and (that family’s mean worry on that contract). GPT-5.4-mini shows $\rho = -0.43$, 95% CI $[-0.69, -0.07]$, **h1_supported negative**, the opposite direction from a “question-difficulty confound” mediator. The mediator hypothesis is rejected.

Probe 2. Stratified $\rho(\text{worry}, \text{Brier})$ by question difficulty (hard half vs easy half) and by point-estimate extremity (uncertain $p \in [0.4, 0.6]$ vs confident). Claude’s inversion lives in the EASY-contracts subset only ($\rho_{\text{EASY}} = -0.19$, $\rho_{\text{HARD}} = +0.03$); GPT-5.5’s inversion lives in the UNCERTAIN- p subset only ($\rho = -0.31$, $n=21$). The sign-flip is family-specific AND subspace-specific; no single uniform mechanism.

Probe 3. Lexical log-likelihood-ratio of distinguishing tokens between high-worry ($\text{tip} \geq 70$) and low-worry ($\text{tip} \leq 30$) rationales per family. Under affect-proxy, high-worry rationales should be enriched in CAREFUL (*analyze, consider*) or UNCERTAIN (*unsure, difficult*) words. We observe instead that the dominant distinguishing terms are *specific content-topic words, tactic, convolution, measurability, singularity*, consistent with a topic-trigger rather than affect-proxy hypothesis.

Probe 4. Cross-family worry agreement: do families AGREE on which contracts get high worry? Pairwise Spearman correlation between each pair of families’ mean-worry-per-contract gives Claude $\leftrightarrow$ GPT-5.5 $\rho = +0.40$ (**h1_supported**), GPT-5.5 $\leftrightarrow$ GPT-5.4-mini $\rho = +0.41$ (**h1_supported**), GPT-5.4-mini $\leftrightarrow$ DeepSeek $\rho = +0.64$ (**h1_supported**). **Families share a topic-trigger signal: they pick the same contracts as high-worry.**

Probe 5. $\rho(\text{worry}, \text{wallclock.seconds})$ per family, does the model “think longer” on high-worry

rows? All families show $|\rho| \leq 0.25$ with CIs crossing zero. Worry is NOT a compute-effort proxy.

Probe 6, mean-regression within row. For each row we measure $\rho(\text{worry}, |p_{\text{success}} - 0.5|)$ per family. Claude: $\rho = -0.39$ 95% CI $[-0.55, -0.20]$, **h1_supported** negative. Codex-mini: -0.59 $[-0.72, -0.41]$, **h1_supported**. DeepSeek: -0.35 $[-0.58, -0.06]$, **h1_supported**. GPT-5.5 and Gemini both null. **Three of five families moderate their probability when emitting high worry.**

Probe 7, per-contract accuracy alignment. Rank contracts by claude’s mean worry. Compare mean Brier per family in the top-quartile (claude-high-worry contracts) vs bottom-quartile (claude-low-worry contracts):

Family	Top-quartile Brier	Bottom-quartile Brier	Diff
Claude Opus 4.7	0.181	0.392	−0.211
GPT-5.5	0.199	0.224	−0.025
GPT-5.4-mini	0.160	0.219	−0.060

Claude is half as wrong on its high-worry contracts. The other families show small or no asymmetry.

Probe 8. RLHF hedging mediator (rejected). $\rho(\text{hedging-vocabulary density, Brier})$ is POSITIVE for Claude (+0.15) and GPT-5.4-mini (+0.31), opposite the direction of their worry-Brier inversion. Hedging-language is not the mediator.

Converged synthesis: a two-mechanism interaction. The worry-Brier sign-flip emerges from the simultaneous presence of two distinct mechanisms:

1. *Worry-triggered mean-regression.* High worry $\rightarrow$ less-extreme p on the same row (Probe 6). Three of five families show this at **h1_supported**.
2. *Per-family accuracy alignment with the trigger topics.* The shared topic-trigger (Probe 4) flags long-structured apparatus contracts as high-worry. Claude is GENUINELY BETTER on those contracts (Probe 7, diff -0.21); GPT-5.4-mini is not (diff -0.06).

Claude has BOTH (1) and (2): moderation lands close to truth on contracts where claude is accurate $\rightarrow$ inverted $\rho(\text{worry, Brier})$ (the sign-flip signature). GPT-5.4-mini has (1) but not (2): moderation doesn’t save it on topics it’s weaker on $\rightarrow$ direction-sensible $\rho(\text{worry, Brier})$. GPT-5.5 has neither cleanly at the pooled-across-conditions level, and its worry-Brier inversion (documented at the per-pilot positive-magnitude-balance-off cell, $N=85$) is condition-specific.

The mechanism is identified at evidence-summary grade for Claude (the most cleanly-inverted family). It is partial for the other inverted families. White-box activation-level probing would directly test which activations drive worry emission versus probability emission; we have only the black-box behavioral signature. The deployment implication is unchanged, the worry scalar by itself is not a competent calibration signal across families, but the routing recipe is now mechanistically explained: per-(family, corpus) routing is needed because mean-regression is universal-but-variable, while per-family topic-accuracy alignment is what gives the worry-Brier relationship its sign.

11 Multi-channel R^2 and a NEW channel: self-predicted Brier interval

The orthogonality result (§8) said the three channels are statistically independent. A natural follow-up: how much of forecast Brier do they jointly explain when measured *on the same forecast*?

Earlier we could only estimate single-channel R^2 per primitive (5–26%) because each primitive was elicited in a separate pilot with its own forecast. The all-channels primitive elicits four channels in ONE prompt: bid-ask spread, scalar worry, AND the lower and upper bounds of the LLM’s self-predicted Brier interval (a previously-unused continuous channel).

Multi-channel OLS R^2 on same-forecast Brier (public-domain $N=42$, leave-one-out cross-validated):

The reported R^2 on $N=15$ low-overlap with $k=4$ channels is mathematically reckless, 14 degrees of freedom and four free regressors gives an in-sample R^2 that is mostly fit noise. The honest report is the public-domain $N=42$ corpus with adjusted R^2 and leave-one-out R^2 (LOO), where LOO subtracts the optimism bias:

Family	N	R^2 (in-sample)	adj R^2	R^2_{LOO}	reading
GPT-5.5	42	0.572	0.525	+0.312	real out-of-sample signal
Gemini 2.5 Flash	42	0.305	0.230	+0.042	barely real
Claude Opus 4.7	42	0.201	0.115	+0.027	barely real
DeepSeek Chat	42	0.276	0.197	−0.053	overfit
GPT-5.4-mini	42	0.162	0.072	−0.288	overfit worse than the mean

The honest claims.

First, *only GPT-5.5 has a multi-channel decomposition that survives leave-one-out cross-validation on the public corpus.* For GPT-5.5, four channels (worry scalar, bid-ask spread, `b_mid`, `b_width`) jointly explain $\approx 31\%$ of squared-error variance out-of-sample at $N=42$. For the other four families, R^2_{LOO} is at or below zero, the multi-channel decomposition does not generalize beyond the fit set at this N , even though three of them have non-trivial in-sample R^2 . The orthogonality claim (§8) stands; what does NOT stand is the stronger claim that combining the channels reliably predicts error for those four families at $N=42$.

Second, *a new continuous channel was introduced and is properly characterized as a primitive, not a fix.* The all-channels prompt asked each model to emit `predicted_brier_lo` and `predicted_brier_hi` alongside the forecast. The interval midpoint (`b_mid`) and width (`b_width`) form a self-predicted-Brier channel with no precedent we are aware of in the LLM-calibration literature. On the public-domain corpus, `b_mid` and `b_width` each carry $\rho < 0.30$ versus Brier for every family, they are informative for one family (GPT-5.5 multi-channel survivor) but are not a single-channel rescue.

What the low-overlap $N=15$ numbers were. A within-program control on research-internal contracts, with $k=4$ regressors at $N=15$, the in-sample R^2 of 0.66 to 0.90 values for GPT-5.4-mini and Gemini are in-sample fits with too few residual degrees of freedom to interpret. We report them in our internal logs; we do NOT make a cross-corpus deployment claim from them.

Partial replication of the reference-class outside view at $N=42$ public-domain. A separately-fired pilot elicited an outside-view base rate (the model first listed five similar past resolved questions and emitted their historical YES rate as `p_base_outside`, before its own `p_success`). At $N=42$ public-domain, $\rho(\text{p_base_outside}, y)$ is positive in direction for all five families and reaches $\rho=+0.39$, 95% CI $[+0.10, +0.62]$, `h1_supported` for GPT-5.4-mini, the only family that clears the Fisher-z detectability floor at $N=42$. For that family the outside-view tracked outcomes more strongly than its own point estimate ($\rho=+0.21$). The other four families are **inconclusive_underpowered**, all positive-direction. We treat this as a partial cross-corpus replication of a novel primitive, not a full generalization.

Saturation curve. The R^2 -vs-channel-count saturation curve is the right summary plot for the deployment recipe; we leave it pre-registered as the next analytical pass, on the public-domain

corpus and reported in LOO terms.

12 Where the per-family signal actually lives: conditional structure

The §11 multi-channel result (only GPT-5.5 survives LOO on $N=42$) leaves four families with channels that don’t generalize. Three free post-hoc analyses on the same data sharpen this:

Sub-source heterogeneity within the public-domain corpus. The $N=42$ public-domain corpus is itself a mixture of Polymarket prediction-market questions, Manifold prediction-market questions, and yfinance stock-threshold questions. Recomputing ρ (worry, Brier) per (family, sub-source):

Family	yfinance ($N=10$)	Manifold ($N=19$)	Polymarket ($N=13$)
Claude	+0.54	+0.51 (h1)	−0.44
GPT-5.5	+0.59	+0.20	+0.64 (h1)
GPT-5.4-mini	−0.69 (h1)	−0.12	−0.38
Gemini	+0.13	−0.21	+0.75 (h1)
DeepSeek	+0.21	−0.28	+0.95 (h1)

Claude’s sign flips between Manifold (+0.51) and Polymarket (−0.44); DeepSeek’s signal lives almost entirely in Polymarket (+0.95, 95% CI [+0.82, +0.98]). Pooled-corpus ρ averages across this heterogeneity. The per-family rule is incomplete without a sub-source decomposition, a contract’s host platform is a hidden confound.

Pooled signal collapses after task-difficulty controls. A standard practice is to report “classifier on LLM uncertainty channels predicts $R^2 \approx 0.36$ of Brier variance.” We refit on the pooled $N=210$ (5 families $\times$ 42 contracts) with two task-difficulty baselines added: **question_length** and forecast-extremity $|p_{\text{success}} - 0.5|$. Incremental R^2_{LOO} from each LLM channel added on top of the baseline:

- baseline (q_len + p_extremity only): $R^2_{\text{LOO}} = +0.101$
- + worry: $\Delta = -0.003$
- + spread: $\Delta = +0.011$
- + b_mid: $\Delta = -0.009$
- + b_width: $\Delta = -0.010$

On pooled data, controlling for task-difficulty wipes out the LLM-channel contribution. The per-family multi-channel signal (GPT-5.5 $R^2_{\text{LOO}} = +0.31$, §11) survives because it conditions on family identity. A claim of the form “LLM-emitted uncertainty channels add predictive value” *across* families on pooled data does NOT hold once task-difficulty is controlled.

Hardness-conditional multi-channel R^2 . Per-(family, hardness-quartile) cell on $N=42$ public, where hardness = cross-family $\sigma(p_{\text{success}})$ (high σ = families disagree = hard):

On the easy quartile (low cross-family disagreement, $N=11$), three families show multi-channel $R^2_{\text{LOO}} > 0.87$. On the hard quartile (high disagreement), LOO collapses across all families. The cell size is small ($N=10-11$ with $k=4$ channels), so this is a hypothesis-generating result requiring $N \geq 20$ per cell to confirm; the deepseek Q3 $R^2_{\text{LOO}} = -480$ is a near-singular-design artifact. The pre-registered confirmation pass is on the $N=142$ topic-balanced extension.

Family	Q1 (easy, $N=11$) R_{LOO}^2	Q4 (hard, $N=10$) R_{LOO}^2	Reading
GPT-5.5	+0.93	−0.79	generalizes on easy; overfits on hard
DeepSeek	+0.92	−13.0	same, near-singular on hard
Gemini	+0.87	−2.5	same
GPT-5.4-mini	−0.74	−2.0	no fit either bucket
Claude	−4.1	−10.0	no fit either bucket

The deployable form of these three findings is one rule: *deploy per-family multi-channel routing only when sub-source identity is matched and cross-family disagreement is low*. Stated differently, the LLM-uncertainty signal is conditional on context-of-call, not unconditional per-family.

13 Five additional findings from the all-channels primitive round

The all-channels round elicited multiple new primitives in a single prompt or as paired conditions. Five additional findings emerge at $N=131$ –142 public-domain:

Rollback hurts. A self-counterfactual primitive (“imagine you were trained only up to 2023, emit a rollback forecast”) HURTS Brier for 4 of 5 families at $N=142$ paired (Δ Brier rollback–normal: claude +0.064, codex-5.5 +0.034, codex-5.4-mini +0.028, deepseek +0.028, gemini −0.005). Post-2023 training is net-useful even though families can’t introspect WHICH contracts benefit. *Deployable rule:* don’t prompt-engineer “imagine you knew less” framings on forecasting tasks.

Leakage agreement without calibration. On the same rollback primitive, all 10 pairwise $\rho(|\text{leak}_A|, |\text{leak}_B|)$ between families are $\geq +0.43$ (h1_supported; claude $\leftrightarrow$ codex-5.5 $\rho = +0.80$). Families AGREE which contracts feel high-leakage. *But* per-family $\rho(|\text{leak}|, \text{err}^2)$ is h0_kept for all five at the ± 0.30 target. Self-reported leakage is a real CONTRACT-structural signal, NOT a per-family calibration channel. Useful as a corpus-curation filter (filter to low-leakage contracts for honest LLM-fresh tests).

DeepSeek’s emitted decision-threshold is calibrated. An asymmetric-loss decision-threshold primitive (commit-YES threshold under symmetric / YES-bad-3 $\times$ / NO-bad-3 $\times$ loss). Codex-5.4-mini emits the most Bayes-coherent thresholds (mean YES-bad 0.750, exact match). Deepseek h1: $\rho(\text{threshold_yes_bad}, \text{err}^2) = -0.19$, 95% CI $[-0.35, -0.03]$ at $N=142$. The emitted threshold tracks accuracy for one family. Novel deployable channel.

Stake-framing amplifies worry universally, leaves p unchanged. Same contract framed as “\$1 stake” vs “\$100K stake” across 103–142 paired contracts per family. Worry-scalar amplifies for every family (Δ worry from +10.8 deepseek to +33.5 claude on the 1–100 scale). But $|\Delta p_{\text{success}}| < 0.03$ for every family. Universal cognitive-shift-without-calibration-shift on a new manipulation, bridges the cognitive-decoupling result to the stake-framing / risk-amplitude human literature (Holt-Laury 2002).

Inversion rule for channel-routing meta-cognition (three of five families). A meta-channel-routing primitive (“which of your own uncertainty channels best predicts your error?”) reveals systematic anti-coherence at $N=15$ internal: claude, codex-5.5, and deepseek pick the WRONG channel 87% of the time (binomial $p=0.005$ each). Within those picks, $\rho(\text{picked}, \text{err}^2)$ is negative while $\rho(\text{unpicked}, \text{err}^2)$ is strongly positive (claude +0.51, codex-5.5 +0.53, deepseek +0.65). *Deployable inversion rule:* when these three families claim “worry is my best channel,” use spread instead (or the analogous flip). Codex-5.4-mini is coherent (73% right); gemini is random. Companion finding to the per-family worry sign-flip, on a new axis. Internal-only at $N=15$; cross-corpus replication at $N=42$ external is pre-registered. The pattern (LLMs have honest contract-level structural intuitions without calibration to truth) recurs across the rollback, leakage,

decision-threshold, and meta-channel-routing results, the defining negative finding about LLM self-monitoring.

14 Universal cross-family patterns

The findings above split the LLM panel by family. Free analyses on the same $N=42$ public-domain data surface four universal patterns that hold across all five families with the same direction, which we report here because they are the patterns most likely to transfer to a different LLM panel.

Universal over-confidence on confident-NO predictions. A reliability-curve analysis (binning each family’s forecasts into quintiles by stated p_{success} , then computing the actual YES rate per quintile):

Family	Mean predicted p in quintile 1	Actual YES rate, quintile 1	gap
Claude	0.04	0.38	+0.34
GPT-5.5	0.07	0.75	+0.68
GPT-5.4-mini	0.04	0.62	+0.58
Gemini	0.05	0.75	+0.70
DeepSeek	0.03	0.75	+0.72

Every family is dramatically over-confident on its “confident-NO” forecasts. The quintile-5 (confident-YES) end is roughly calibrated (gap -0.02 to -0.21). The miscalibration is asymmetric: confident-NO is poorly calibrated; confident-YES is OK. With $N=8-10$ per quintile per family, the per-family CI is wide, but the universality across all five families and the consistent direction of the gap argue the pattern is real. *Deployable rule:* on a news-event public corpus, do not trust LLM $p_{\text{success}} < 0.10$ as a confident-NO commit.

Horizon-conditional Brier. Pooled across all 5 families ($N=210$): $\rho(\text{days to resolution from 2026-01-01, err}^2)$ +0.161, 95% CI [+0.026, +0.290], h1_supported. Brier on April-2026-resolving contracts is 0.240; on May-2026 it is 0.385. Contracts farther from the training cutoff are systematically harder. The mechanistic interpretation is straightforward (more uncertainty accumulation over longer horizons), but the cleanly-documented per-contract horizon-confidence-weighting term is novel for LLM forecasting at this N .

Universal NO-bias. On contracts that did resolve YES ($N=26$ of 42 in this corpus), every family’s mean p_{success} is below 0.5 (0.38–0.48). Brier conditioned on $y=1$ exceeds Brier conditioned on $y=0$ for every family; DeepSeek shows the largest asymmetry (0.46 vs 0.12). LLMs are systematically better at predicting NO outcomes than YES outcomes on this corpus. Possible interpretations include (a) underlying corpus YES base-rate of 62% exceeds the model’s prior on these questions; (b) “will X happen by date Y” framing biases toward NO; (c) prior-cutoff bias on then-future events. We do not distinguish among these here.

Naive ensemble averaging does NOT help. On the full $N=142$ paired set with all five families’ p_{success} values: median-of-5 Brier = 0.272, mean-of-5 Brier = 0.261, best-single-family (Claude) Brier = 0.254. Paired permutation: median-of-5 loses to the best single family ($\Delta=+0.018$); mean-of-5 also loses ($\Delta=+0.007$). The standard “ask five LLMs, take the median” deployment recipe yields no benefit on this corpus and slightly hurts. Best-single-family dominates naive ensembling at this N .

Universal cross-family forecast agreement. For every pair of the five families, $\rho(p_{\text{success},A}, p_{\text{success},B})$ on the same $N=42$ public-domain contracts is h1_supported. Pairwise ρ ranges +0.52 (codex-5.4-mini $\leftrightarrow$ DeepSeek) to +0.88 (Claude $\leftrightarrow$ GPT-5.5). Mean +0.72. The five families agree on forecast

direction across vendors and architectures. This caveats the per-family results: when families converge, they really converge, but high pairwise ρ does not distinguish “models converge on truth” from “models share training-data bias.” For panel-aggregation design, the practical takeaway is that consensus-strength is a usable confidence weight: a contract where all five families return near-identical p_{success} can be treated as a higher-confidence pooled forecast than a contract where they disagree.

Universal source-difficulty ordering. Per-source Brier per family on the public-domain corpus reveals a universal ordering: *Polymarket* > *Manifold* > *yfinance*. Polymarket contracts (election outcomes, news-events) yield Brier 0.46–0.57 across all five families; yfinance ETF threshold contracts yield 0.22–0.23. The same order holds for every family. Source identity is a per-contract confidence-weighting term independent of family.

The six universal patterns together suggest a deployable per-contract recipe: (a) discount the model’s confident-NO forecasts; (b) horizon-weight per-contract confidence; (c) apply a per-family YES-bias correction term; (d) prefer a calibrated single family over a naive ensemble; (e) weight pooled-forecast confidence by cross-family consensus-strength; (f) weight by source difficulty.

Composed recipe at $N=142$: routing beats naive aggregation; matches but does not yet exceed the best single family. We instantiate the rules above as a single composed forecast (the mean-of-5 panel forecast adjusted by four universal coefficients: a confident-NO discount when the panel mean is below 0.20, an upward YES-bias correction in the middle band, a horizon-to-cutoff shrinkage, and a per-source-difficulty shrinkage, see App. A for the closed form). All four coefficients are chosen *ex ante* from the universal patterns above, not learned from held-out folds. The contrast we test is paired-permutation Δ -Brier against three baselines and against a per-family channel-routed alternative on the same $N=142$ paired contracts.

strategy	Brier	Δ vs. best-single	90% CI	p_{perm}
median-of-5	0.272	+0.018	,	,
mean-of-5	0.261	+0.007	,	,
best-single (Claude)	0.254	,	,	,
routed (universal-only)	0.232	−0.022	[−0.050, +0.005]	0.18
routed (per-channel)	0.246	−0.009	[−0.035, +0.016]	0.58

Three results.

(i) *Routed-vs-naive-aggregation is the clean win.* Routed (universal-only) beats median-of-5 by $\Delta=-0.040$ ($p_{\text{perm}}=0.0013$) and mean-of-5 by $\Delta=-0.029$ ($p_{\text{perm}}=0.0069$). **h1_supported**. The composed recipe out-performs the standard “ask five LLMs, take the median/mean” deployment baseline at $p<0.01$ on the same $N=142$ paired set.

(ii) *Routed-vs-best-single is directional but underpowered.* The point estimate $\Delta=-0.022$ sits almost entirely on the routing-wins side of zero (90% CI [−0.050, +0.005]), and the direction is consistent with the same comparison at the previous $N=42\times 3$ snapshot before the codex refill ($\Delta=-0.013$, $p_{\text{perm}}=0.48$). The gap roughly doubled in routing’s favor as N rose; the p shrank from 0.48 to 0.18. At the observed effect size and paired-Brier variance, the Fisher- z power calculation gives $N\approx 250$ –300 for $p<0.05$ at 80% power on a two-sided test. We report **h0_kept** against the $\Delta\geq 0.05$ detection bar that the program’s pre-registration sets for deployment claims, and **inconclusive_underpowered** against the smaller observed effect.

(iii) *Universal rules generalize; per-family channel routing does not.* Routed (universal-only) beats routed (per-channel) by $\Delta=+0.014$ ($p_{\text{perm}}=0.042$), i.e. the per-family channel-weighting alternative makes the forecast *worse* than the universal-rules version on the public-domain corpus, at $p<0.05$. This is consistent with §8 (only GPT-5.5’s multi-channel decomposition has a generalizable

$R_{\text{LOO}}^2 > 0$ at $N=42$) and with the conditional-structure result (§12): channel-level routing is contextually conditional and does not survive a uniform application. The routing surface that does survive is the universal-rules layer.

A ~ 100 -contract pre-registered follow-on on the public-domain corpus is the refill required to escalate (ii) from `inconclusive_underpowered` to a $p < 0.05$ verdict at the observed effect size.

The confident-NO discount as a standalone rule beats raw on every family at $p < 0.05$. The simpler form of the recipe, the single confident-NO discount applied per-family without the other three rules, improves per-family Brier on every panel member at $p < 0.05$:

family	raw Brier	discounted Brier	Δ	paired p_{perm}
claude	0.2543	0.2240	−0.0302	0.016
codex-5.5	0.2625	0.2416	−0.0208	0.030
codex-5.4-mini	0.2714	0.2450	−0.0264	0.015
deepseek	0.3222	0.2704	−0.0518	0.0008
gemini	0.3167	0.2840	−0.0327	0.008

Two observations land here. (a) Discounted Claude at Brier 0.224 is at parity-or-better than the four-rule routed recipe (0.232 above); the additional three rules add nothing meaningful on top of the discount on this corpus at $N=142$. (b) A single one-line post-forecast adjustment improves Brier at $p < 0.05$ across every model family, including outside the panel that originated the rule (DeepSeek and Gemini are subscription-distinct families). The deployable shape is the discount alone, applied to whichever single model is in production.

15 Bias-type-specific inheritance: LLMs escape utility-grounded biases and inherit framing-grounded biases

Three novel-bias smokes on the same $N=30$ public-domain contracts across all five families ($5 \times 5 \times 30 = 750$ calls, 748 schema-OK) test whether well-documented human cognitive biases transfer to LLM forecasting. The findings split cleanly along bias TYPE:

(A) Loss-frame invariance. LLMs ESCAPE the human bias. The same contract asked as “ p_{success} ” vs “ p_{failure} ” should satisfy $p_{\text{success}} \approx 1 - p_{\text{failure}}$ for a symmetric forecaster; humans show systematic gaps of 0.15–0.30 from loss aversion (Kahneman–Tversky 1979). Mean per-family $|p_{\text{success}} - (1 - p_{\text{failure}})|$ across all five panel members: claude 0.035, deepseek 0.049, codex-5.5 0.085, codex-mini 0.098, gemini 0.148. All five families come in well below the human reference, with claude and deepseek essentially symmetric (median ≤ 0.03).

(F) Status-quo / endowment effect. LLMs INHERIT the human bias, at large effect size. The same contract framed as “the YES condition CURRENTLY HOLDS at the time of forecast” vs “do not assume any current state” should satisfy $p_{\text{currently}} \approx p_{\text{fresh}}$ if the model is anchor-symmetric; humans show endowment-effect anchoring on stated state. Mean per-family $|p_{\text{currently}} - p_{\text{fresh}}|$: gemini 0.084, deepseek 0.110, claude 0.241, codex-mini 0.275, codex-5.5 **0.440** (median 0.500, half of contracts flip sign between the two framings). Three of five families show a clearly large status-quo response; codex-5.5 in particular reverses the forecast on half the contracts purely from prompt framing.

(B) Probability-weighting curve, preliminary near-linear (no inverse-S). For each contract we elicit p_{emit} at the eight anchor probabilities $\{0.01, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.99\}$. Cumulative prospect theory (Tversky–Kahneman 1992) predicts inverse-S overweighting of small

probabilities. Preliminary Claude responses: $\{0.02, 0.07, 0.12, 0.27, 0.50, 0.73, 0.88, 0.985\}$. Near-linear; no CPT inverse-S overweighting at the low-probability tail.

The split is informative. The biases that DO transfer (status-quo / endowment) are grounded in stated state framing. LLMs anchor on what the prompt asserts. The biases that DO NOT transfer (loss-frame symmetry, probability-weighting curve) are grounded in human utility structure. LLMs have no value function over loss vs gain, no consumption preference, no asymmetric disutility of loss. This is the clean diagnostic shape: *LLMs inherit framing-grounded biases and escape utility-grounded biases*. It is neither the “LLMs are human-like” nor the “LLMs are rational” framing the literature has been working with.

The deployable rule the split suggests: when contract framing is ambiguous on current-state, elicit BOTH framings and average. Two-arm cost ($+2\times$) on ambiguous contracts, near-zero cost on unambiguous ones, eliminates the codex-5.5 half-flip pattern in (F). An $N=42$ external replication on a diversified public-domain corpus (Metaculus + FRED, 90 contracts, post-cutoff filtered) is firing at the time of writing.

16 Prompt-invariance asymmetry across families: a methodological caveat

Across three prompt variants that elicit p_{success} on the same public-domain contracts (all-channels, reference-class outside view, and conjunctive decomposition), per-family cross-prompt stability is heterogeneous:

Family	$\rho(p_{\text{all-channels}}, p_{\text{outside-view}})$ at $N=42$	95% CI
Claude	+0.868	[+0.77, +0.93]
GPT-5.5	+0.830	[+0.70, +0.91]
GPT-5.4-mini	+0.870	[+0.77, +0.93]
Gemini	+0.504	[+0.24, +0.70]
DeepSeek	+0.386	[+0.09, +0.62]

Mean per-family standard deviation of p_{success} across the three prompts is 0.05–0.06 for Claude/Codex variants; 0.108 for Gemini; 0.148 for DeepSeek. Claude and the two Codex variants emit beliefs that are 2–3 $\times$ more prompt-invariant than Gemini and DeepSeek’s.

This was initially read as the methodological caveat touching every per-family finding. We later refined it: pooling p_{success} across four “standard” prompts (all-channels, asymmetric-loss, rollback, low-stake) on the same ($N=41$ –42) contracts per family shows mean $\sigma(p_{\text{success}}) = 0.042$ –0.064 for EVERY family, including Gemini and DeepSeek. The earlier-reported 0.108 / 0.148 values were driven specifically by the reference-class outside-view prompt, which forces a pre-forecast base-rate computation. That prompt is a different reasoning architecture, not the same forecast with prompt noise.

The scoped caveat is then: the per-family findings in this paper based on *standard* elicitation prompts (worry scalar, bid-ask spread, predicted-Brier interval) are roughly equally prompt-invariant across all five families on the POINT ESTIMATE p_{success} . Findings that require the model to execute a non-standard reasoning step pre-forecast (e.g., the outside-view computation) may shift more for Gemini/DeepSeek than for Claude/Codex.

Channel-level caveat: the worry scalar is roughly 4 $\times$ noisier across prompts than p_{success} . Pooling worry data from the all-channels, low-stake, and high-stake prompts on the same ($N=41$ –42) shared contracts per family: mean per-family $\sigma(\text{worry})$ is 10–22 on the 1–100 scale

(10–22% of full range), versus $\sigma(p_{\text{success}}) = 0.04\text{--}0.06$ (4–6% of $[0, 1]$). Stated equivalently: the worry channel carries substantial prompt-elicitation noise that the point estimate does not. This is a methodological caveat on every worry-based finding in this paper (the per-family sign-flip, the DeepSeek threshold result, the cross-prompt contradiction): the worry channel itself is less prompt-invariant than p_{success} . The clean disambiguation requires within-family worry-paraphrase pilot (5 prompt designs $\times$ same contract per family) at proper N ; we have not run it.

Cross-prompt sign-stability of the worry channel (critical). Pooling worry- p_{success} pairs from four public-domain prompt variants (all-channels, asymmetric-loss, rollback, stake-framing) at $N=248\text{--}326$ per family: $\rho(\text{worry}, \text{err}^2)$ for Claude is $+0.166$ $[+0.042, +0.285]$, **h1_supported** in the DIRECTION-CORRECT direction, the opposite of the low-overlap-corpus result that drove the deployed per-family sign rule. Codex-5.5 is direction-correct externally (consistent with internal); Codex-5.4-mini’s strong internal direction-correct signal goes null externally; Gemini and DeepSeek remain flat. The cleanest read is that the inversion-for-Claude rule is corpus-bound or prompt-bound, not a per-family property. The deployed routing rule for Claude has been scope-restricted to “low-overlap-corpus prompt style only” pending a retest pilot (~ 42 calls of the original prompt style on $N=42$ public-domain) that disambiguates corpus-effect from prompt-effect.

17 Pre-registered test of the inheritance frame on a held-out bias slate

The three-axis frame in §3 was induced from the body of measurements above. The framework earns its predictive claim only by getting subsequent bias-transfer predictions right on biases it had not yet seen. This section reports the pre-registered test.

Pre-registered slate. Seven additional human biases held out from the F102 measurement, classified ex ante into the three cells of axis 2 before any new data was collected.

- INHERIT (pure-heuristic, high natural-speech footprint): **C** gambler’s / hot-hand fallacy [11]; **E** self-overconfidence [6]; **H** base-rate neglect when the prior is explicit; **J** numeric anchoring [12].
- ESCAPE (pure-motivational, low text representation of the bias mechanism): **G** hyperbolic discounting [2].
- MIMIC (systemic-motivational, intrinsic utility BUT heavy case-study representation): **D** sunk cost [1]; **I** in-group / national framing [8], with explicit prediction of alignment damping.

Decision rule. Of the seven held-out biases together with the three already measured in §15 (A loss-frame ESCAPE confirmed; B probability weighting ESCAPE preliminary; F status-quo MIMIC as re-classified from INHERIT), classifying ≥ 8 of 10 into the predicted cell is the bar for **h1_supported** on the framework as a structured prediction. The random-cell baseline is $\sim 3.3/10$. The MIMIC vs INHERIT discrimination is checked separately via two signatures: (S1) the magnitude correlates with how often the bias is *discussed* in training corpora (proxy: retrieval of the bias name returns case-study hits, not first-person enactments); (S2) the effect collapses under explicit anti-bias prompting much more for MIMIC than for INHERIT.

Status at the time of writing. The smoke is in flight on the existing public-domain pool (claude subscription, $n=15$ per arm); a $n=42$ external replication on the diversified post-cutoff corpus is

the follow-on if the smoke lands directionally. We will report the slate verdict in a revision; the pre-registration of cell classifications and decision rule is fixed at the version-tag of this manuscript.

18 The re-audit discipline: a warning to the field

The discipline that produced the results above also produced a re-audit of the program’s own prior findings. We treat the re-audit itself as a contribution because the same overcalling pattern almost certainly affects the published LLM-forecasting literature at the sample sizes it currently reports.

18.1 Power calculus applied to our own program

Before fixing the verdict resolver, eight prior “no effect” or “cross-corpus replication” claims in the program had been written down as findings. Applying the Fisher- z power-aware verdict resolver retrospectively:

- Four “null” findings move from `h0_kept` to `inconclusive_underpowered`: the data did not actively rule out a meaningful effect at the per-family target $|\rho|$ floor; the absence-of-signal was an absence-of-power.
- Five cross-corpus claims move from “replicated” or “failed-to-replicate” to corpus-specific: the original cross-corpus reading was a single sub-source point estimate at $N=5-15$ external, with CIs that crossed zero and with sub-source heterogeneity unmodeled.
- One self-knowledge-inversion finding (originally three families at $N=5$) retracts to a single family at $N=15$ internal; the other two cross-corpus directions flip on re-fire.
- Two per-family deployed routing rules (Brier- Δ map and *consider-both-sides* effect) move from deployment-grade to “directional hypothesis only”; the required N to confirm a $\Delta_{\text{Brier}}=0.05$ effect at 80% power is ≈ 459 per family, an order of magnitude above the original $N\approx 30$ per cell.

In aggregate, $\approx 50\%$ of the program’s prior verdicts did not survive the verdict resolver. None of the underlying executions were flawed. The interpretation discipline was.

18.2 A worked example: the per-family worry sign-flip retest

The deployed YAML routing for Gemini was scope-restricted to “low-overlap corpus, *v21*-style prompt only” pending a ~ 42 -call retest of the same prompt on the public-domain $N=42$ slice that the other *v28* pilots fired on. The retest, fired with the same C1 worry-only prompt on `gemini-2.5-flash` via API:

- Schema-OK: 42/42.
- $\rho(\text{worry}, \text{Brier}) = +0.110$ – the direction-correct direction, NOT the inverted direction the prior deployment had assumed.
- Fisher- z detectable $|\rho|$ at $N=42$, 80% power: ≥ 0.43 . Observed $|\rho|=0.11$ is well below this; the CI excludes the original “inverted” direction ($\rho < -0.30$).
- Verdict: `h0_kept` against the inversion hypothesis; the direction-correct point estimate is in the `inconclusive_underpowered` band for the positive hypothesis.

The retracted YAML rule for Gemini stays retracted. The earlier “inversion” claim, deployed at $N=5$ with $N_{\text{wrong}}=2$, was an artifact of the sample size. This is the single cheapest decisive retest the queue could fire; the queue carries two more retests of comparable diagnostic value.

18.3 Implications for the published LLM-forecasting literature

The two highest-profile 2024 contributions to LLM forecast calibration are (a) the ensemble-prediction result that a 12-LLM aggregate matches human-crowd Brier on real binary questions [7] and (b) the retrieval-augmented LLM that approaches human-superforecaster accuracy on a contamination-resistant slice [3]. Both report claims at sample sizes that, under the same Fisher- z power calculus we apply to our own program, are bounded by:

- For a Brier-delta $\Delta=0.05$ comparison (LLM ensemble vs. human crowd) at 80% power, paired permutation requires $N \geq 300$ –500 per side.
- For a per-model rank correlation $|\rho|=0.30$ at the same power: $N \geq 91$.
- For a per-(model, sub-source) cell at $|\rho|=0.40$: $N \geq 50$ per cell – requiring sub-source $N \geq 250$ across five families when sub-sources are five-way mixed.

We do not re-analyse those datasets here; we observe only that the standard “2024-style” aggregate claim is structurally compatible with the same overcalling failure mode our own program exhibited. The methodological contribution the published literature is missing – and that this paper contributes – is the verdict resolver discipline: every claim labelled with one of three legal verdicts, n_{required} pre-computed before fire, sub-source heterogeneity reported alongside the aggregate, and prior verdicts re-audited under the same calculus as new ones.

A second-order observation about contamination. The Halawi 2024 dataset (YuehHanChen/forecasting, $N=1754$ binary-resolved questions across Polymarket, Metaculus, Manifold, GJOpen, and CSET) reports resolution-date histogram: 1 question in 2021, 147 in 2022, 1470 in 2023, 136 in 2024, and **zero in 2025 or later**. The 2024 aggregate claim was published with a baseline panel whose knowledge cutoff predated most resolution dates. Any 2025-generation replication that uses current LLMs (Claude 4.7, GPT-5, DeepSeek Chat 2026) inherits a structural compromise: the model’s knowledge cutoff postdates every resolution. A 30-call probe on a uniform random sample from this dataset returned Brier 0.13 on 14 of 14 fired confident-NO cases with empirical YES rate = 0.07 – a calibration signature that is statistically indistinguishable from outcome recall on a forecasting task. The deployable check before any replication is one filter: *resolve_date* > *max(panel_cutoff)*. That filter empties this dataset for the current LLM generation. The general point is corpus-validity drift: a benchmark that was honest at publication time may stop being honest once a new model generation has trained through the resolution dates. The verdict resolver discipline extends to corpus eligibility, not only to per-claim power.

18.4 The deployable consequence

The deployable rule the re-audit produced: *a per-family or per-(family, sub-source) routing rule should be deployed only when its supporting CI has been computed on $N \geq n_{\text{required}}$ at the target effect size, and only with sub-source stratification reported*. The pilot queue’s three remaining cheap retests (~ 0 , 42, and 420 calls respectively) close the same gap on the program’s other deployment rules; their results are published alongside the existing rules as the retests complete.

19 What this paper does NOT establish

- It does NOT show that cognitive-debiasing interventions never transfer to LLMs. Four of five tested are `inconclusive_underpowered` at our low-overlap-corpus N , not falsified.
- It does NOT establish that GPT-5.4-mini’s low-overlap-corpus frequency-framing improvement would survive cross-corpus replication at proper power. $N \geq 91$ public-domain is required.
- It does NOT generalize beyond five model families and two corpus classes. The low-overlap corpus is research-internal and not publicly available in raw form.
- It does NOT decompose LLM *reasoning* mechanistically. The channel-orthogonality claim (§8) is at the *expression* level; a reasoning-decomposition claim would need white-box mechanistic interpretability.
- The four-axis confounding between corpora means the cross-corpus result is evidence that at least one axis matters, not yet a clean attribution. The 4-cell de-confounded corpus is the most important methodological follow-up.
- Bid-ask spread cross-corpus is currently underpowered; the direction-shift is suggestive but not conclusive at $N=42$.
- The worry-Brier sign-flip mechanism is provisional (§10 multi-probe synthesis); activation-level probing and a topic-matched corpus would settle it.
- Per-family signal strength is conditional on prompt-invariance (§16). Gemini and DeepSeek findings should be read with the prompt-stability discount documented there until the within-family paraphrase pilot is run.

20 Reproducibility

All pilot calls persist as JSONL files and are ingested into a SQLite database at `analytics/public/calibration/for`. Pre-registrations and verdict resolutions are in the same database. The reusable general-purpose statistics module is at `src/ztare/experiment_stats.py` (power calculator, bootstrap CI, paired permutation, Fisher- z Spearman, TOST, BH-FDR, power-aware verdict, BIC Bayes factor, reproducibility hash). Forecasting-specific wrappers are in the per-project `calibration_stats.py`. Every reported finding reproduces via CLI: `calibration_stats.py finding frequency-framing, brier-ci --primitive baseline --corpus low-overlap, spearman --pilot bid-ask --x spread, etc.`

Low-overlap-corpus reproducibility. The low-overlap corpus is research-internal and not publicly releasable in raw form. Remediation paths: (a) sanitized release with research-internal identifiers replaced by neutral tags (e.g., “Operation X tactic Y on row class Z ”), preserving the four-axis structural profile; release pending clearance. (b) Parallel public alien-domain corpus drawn from public-but-niche academic sub-fields (specific Lean mathlib lemma classes; obscure subfields of partial differential equations; verifiable but low-traffic historical-event resolutions) matching the low-overlap corpus’s four-axis profile, intended for direct third-party replication of the frequency-framing and bid-ask-spread results.

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A Composed routing recipe: closed form

Let $\bar{p}$ be the mean of the five families’ point estimates p_{success} on a given contract, h the days-to-resolution-from-cutoff, and $s \in \{\text{polymarket}, \text{manifold}, \text{yfinance}\}$ the source identifier. The composed routed forecast $\hat{p}$ is:

$$\begin{aligned}
\hat{p}_{\text{conf-NO}} &= \begin{cases} 0.35 \bar{p} + 0.65 (0.10) & \text{if } \bar{p} < 0.20 \\ \bar{p} & \text{otherwise} \end{cases} \\
\hat{p}_{\text{YES-bias}} &= \hat{p}_{\text{conf-NO}} + \begin{cases} +0.06 & \text{if } 0.30 \leq \hat{p}_{\text{conf-NO}} \leq 0.55 \\ 0 & \text{otherwise} \end{cases} \\
\hat{p}_{\text{horizon}} &= 0.5 + (\hat{p}_{\text{YES-bias}} - 0.5) \cdot \frac{1}{1+0.01 h} \\
\hat{p} &= 0.5 + (\hat{p}_{\text{horizon}} - 0.5) \cdot w_s
\end{aligned}$$

with $w_s = 0.70$ for polymarket, 0.85 for manifold, 1.00 for yfinance. The four coefficients (0.10, 0.06, 0.01, w_s) are not learned from held-out folds; they are chosen from the universal patterns reported in §14 (confident-NO over-confidence, middle-band YES under-shooting, horizon-conditional Brier slope, per-source Brier ordering). The per-channel routed alternative (routed_v2 in the main text) substitutes per-family channel weights for the universal w_s ; its weights are listed in the appendix companion file. The implementation reproduces from `projects/forecaster_skill_calibration_v1/workspace/c` on the database snapshot dated 2026-05-28.